

Departement Elektriese, Elektroniese en
Rekenaar-Ingenieurswese

Department of Electrical, Electronic and
Computer Engineering

Eksamen

Kopiereg voorbehou

Examination

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Module EBN122
Elektrisiteit en Elektronika
14 November 2011

Module EBN122
Electricity and Electronics
14 November 2011



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Eksamensinligting:

Exam information:

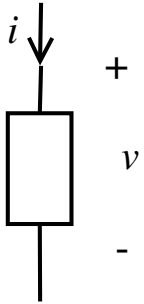
Maksimum punte: <i>Maximum marks:</i>	90	Volpunte: <i>Full marks:</i>	90
Duur van vraestel: <i>Duration of paper:</i>	180 minute <i>180 minutes</i>	Oopboek / toeboek: <i>Open / closed book:</i>	Toe <i>Closed</i>
Totale aantal bladsye (hierdie blad ingesluit): <i>Total number of pages (including this page):</i>		29	

BELANGRIK- IMPORTANT

- Die eksamenregulasies van die Universiteit van Pretoria geld.
The examination regulations of the University of Pretoria apply.
- Alle vrae moet op die ANTWOORDBLAD beantwoord word.
All Questions must be answered on the ANSWER SHEET.
- Studente mag hul eie rowwe berekeninge op die vraestel doen – die vraestel word nie ingeneem nie.
Students may do their own rough calculations on the question paper – the question paper will not be taken in.
- Polêre vorm: $A\angle\theta$; Reghoekige vorm: $a + jb$
Polar form: $A\angle\theta$; Rectangular form: $a + jb$

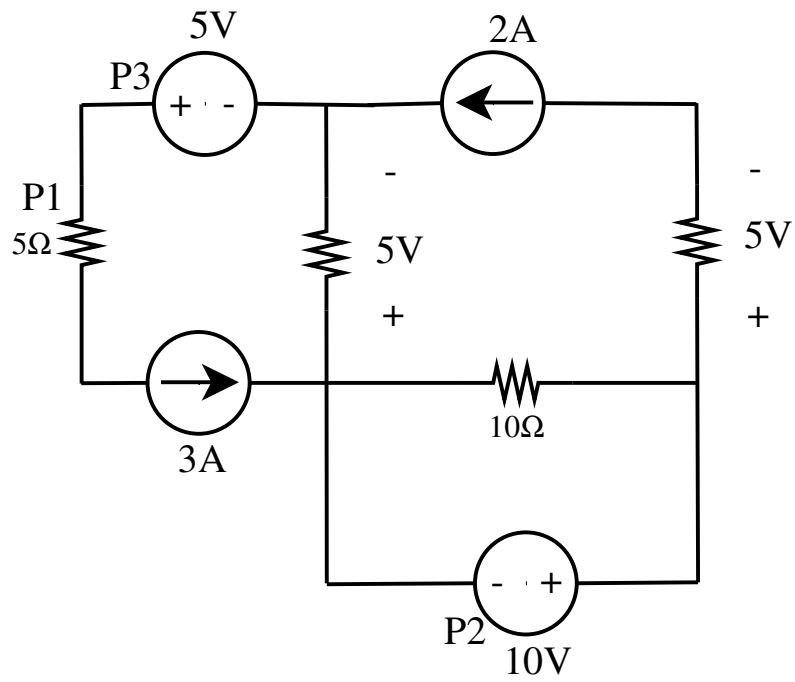
Dosent(e):	1 J Joubert		
Lecturer(s):	2 W Badenhorst		
Groephoof: Ek bevestig dat die vraestel die uitkomstes toets soos gespesifiseer in die studiehandleiding. Group head: I confirm that the question paper evaluates the outcomes as specified in the study guide.			
Groephoof: Group Head:	Prof J Joubert	Handtekening: Signature:	

VRAAG/QUESTION 1 [11]

Vraag 1.1	Question 1.1
	
(a) Die spanning en stroom by die terminale van die stroombaan is nul vir $t < 0$. Vir $t \geq 0$ is dit $v = 2e^{-5t}$ V en $i = 20 - 5e^{-5t}$ A. Bepaal die drywing by $t = 1$ ms. [2] (b) Die spanning en stroom by die terminale van die stroombaan is nul vir $t < 0$. Vir $t \geq 0$ is dit $v = 3 - t$ V en $i = 6 - 4t$ A. Hoeveel energie word aan die stroombaan gelewer tussen 0 en 3 s. [2]	(a) The voltage and current at the terminals of the circuit are zero for $t < 0$. For $t \geq 0$ they are $v = 2e^{-5t}$ V and $i = 20 - 5e^{-5t}$ A. Find the power at $t = 1$ ms. [2] (b) The voltage and current at the terminals of the circuit are zero for $t < 0$. For $t \geq 0$ they are $v = 3 - t$ V and $i = 6 - 4t$ A. Determine the energy delivered to the circuit between 0 and 3 s. [2]

Vraag 1.2

Question 1.2

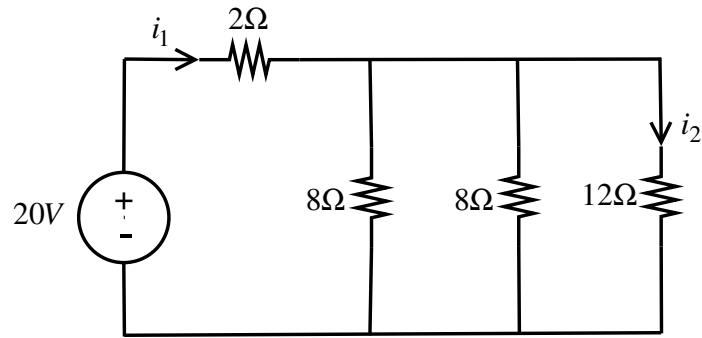


Bepaal die drywings $P1$, $P2$, en $P3$. [3]

Determine the powers $P1$, $P2$, and $P3$. [3]

Vraag 1.3

Question 1.3

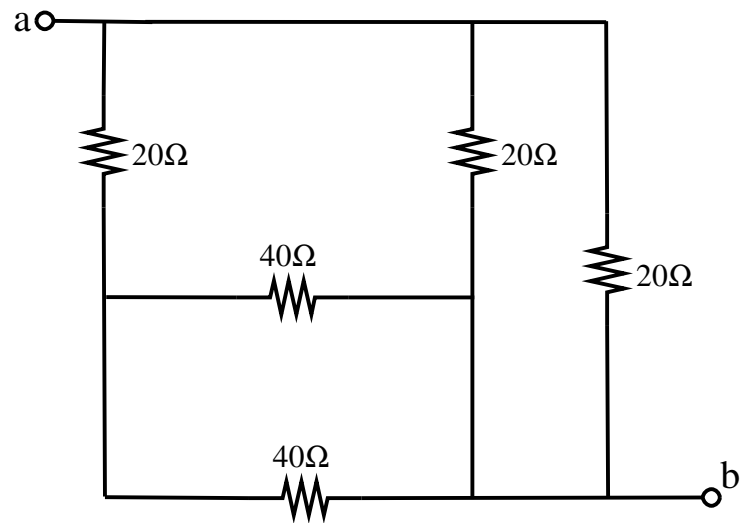


Bepaal i_1 en i_2 in die stroombaan. [2]

Determine i_1 and i_2 in the circuit. [2]

Vraag 1.4

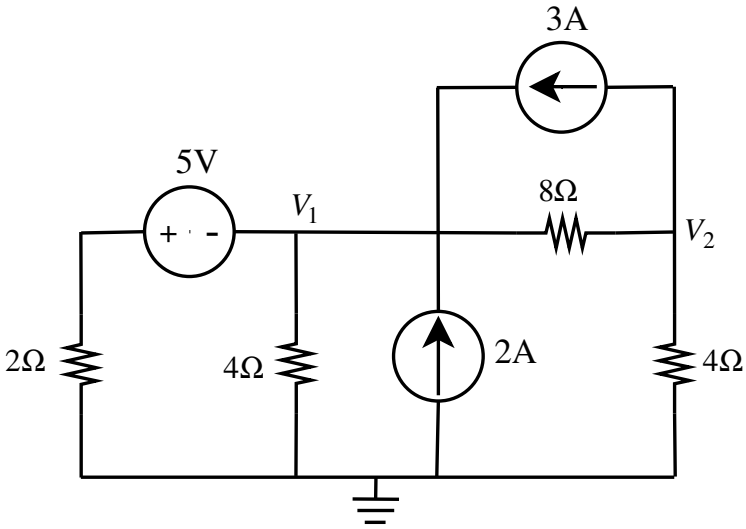
Question 1.4



Reduseer na 'n enkele weerstand tussen terminale a-b. [2]

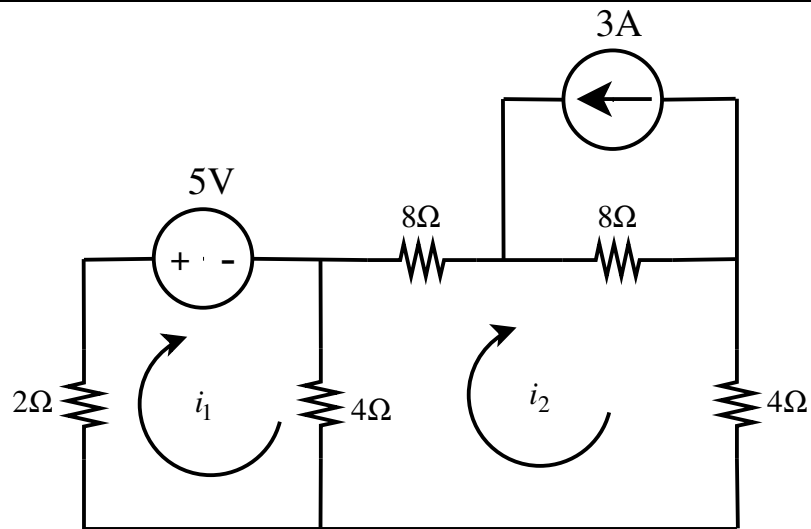
Reduce to a single resistor between terminals a-b. [2]

VRAAG/QUESTION 2 [34]

Vraag 2.1	Question 2.1
	
Gebruik node-analise om twee vergelykings van die volgende vorm op te stel vir die stroombaan: $aV_1 + bV_2 = 20$ $cV_1 + dV_2 = -24$ [4]	Use node analysis to set up two equations of the following form for the circuit: $aV_1 + bV_2 = 20$ $cV_1 + dV_2 = -24$ [4]

Vraag 2.2

Question 2.2



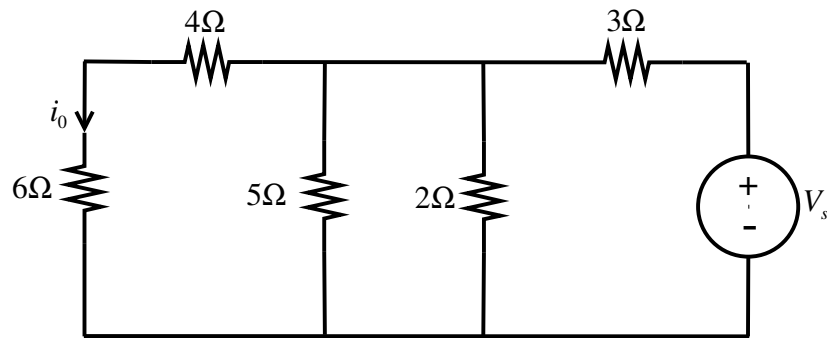
Gebruik lus-analise om twee vergelykings van die volgende vorm op te stel vir die stroombaan:

$$\begin{aligned} ai_1 + bi_2 &= -5 \\ ci_1 + 24i_2 &= d \end{aligned} \quad [4]$$

Use mesh analysis to set up two equations of the following form for the circuit:

$$\begin{aligned} ai_1 + bi_2 &= -5 \\ ci_1 + 24i_2 &= d \end{aligned} \quad [4]$$

Vraag 2.3	Question 2.3
(a) Bepaal V_I. [2] $\begin{bmatrix} 2 & -3 \\ -3 & 2 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$ (b) Bepaal i_2. [2] $\begin{bmatrix} 2 & 3 & 0 \\ 1 & 0 & 2 \\ 4 & 0 & 1 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$	(a) Determine V_I. [2] $\begin{bmatrix} 2 & -3 \\ -3 & 2 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$ (b) Determine i_2. [2] $\begin{bmatrix} 2 & 3 & 0 \\ 1 & 0 & 2 \\ 4 & 0 & 1 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$

Vraag 2.4
Question 2.4


(a) Bereken V_s indien $i_0 = 1\text{ A}$. [2]

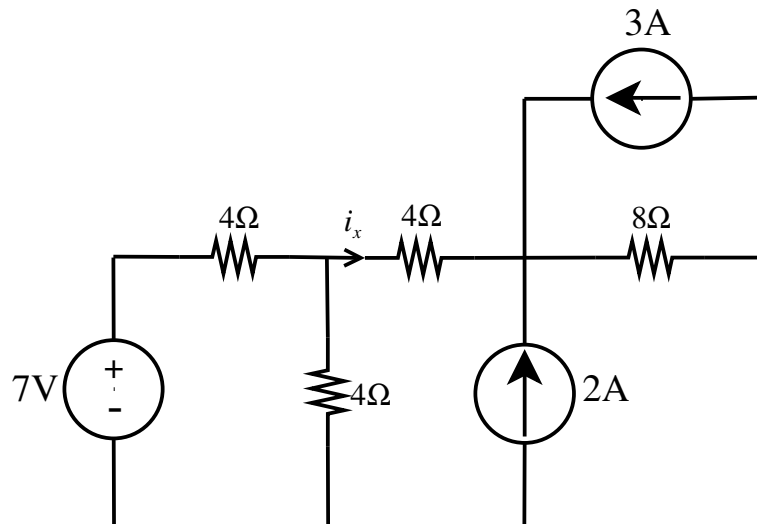
(b) Met watter weerstand kan 'n mens die 6Ω weerstand vervang om te verseker dat optimum drywing in daardie weerstand gedissipeer sal word. [2]

(a) Determine V_s if $i_0 = 1\text{ A}$. [2]

(b) With what value resistor can one replace the 6Ω resistor to ensure that optimum power will be dissipated in that resistor. [2]

Vraag 2.5

Question 2.5

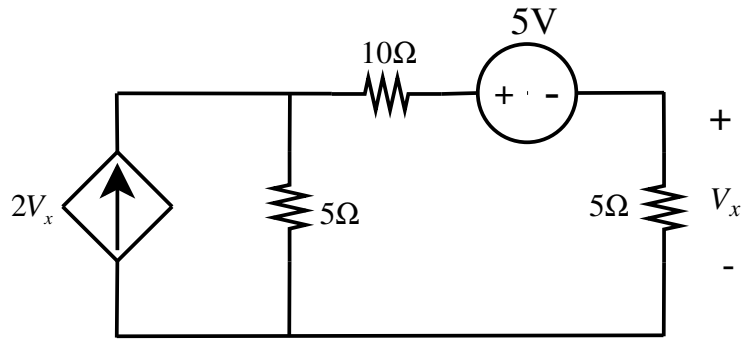


- (a) Bepaal i_x agv die 7V bron. [2]
(b) Bepaal i_x agv die 3A bron. [2]

- (a) Determine i_x due to the 7V source. [2]
(b) Determine i_x due to the 3A source. [2]

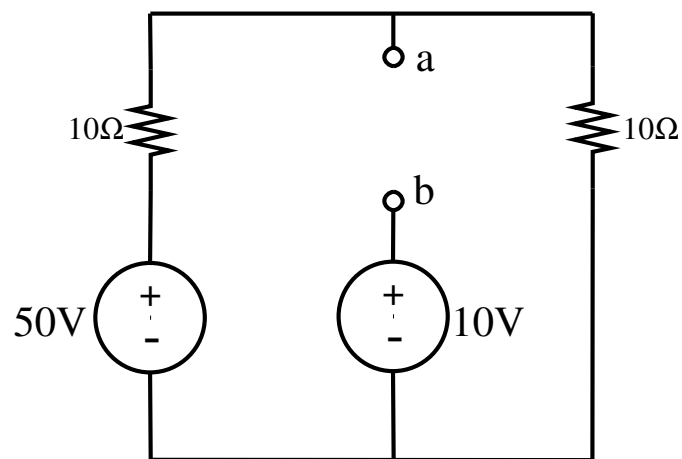
Vraag 2.6

Question 2.6

Gebruik brontransformaties om V_x te bereken. [2]Use source transformations to determine V_x . [2]

Vraag 2.7

Question 2.7

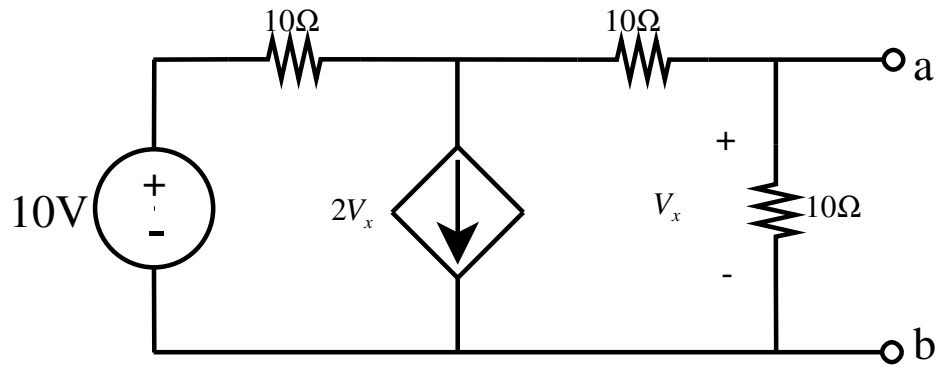


Bereken die Thevenin ekwivalent tov terminale $a-b$.
[4]

Determine the Thevenin equivalent wrt terminals $a-b$.
[4]

Vraag 2.8

Question 2.8

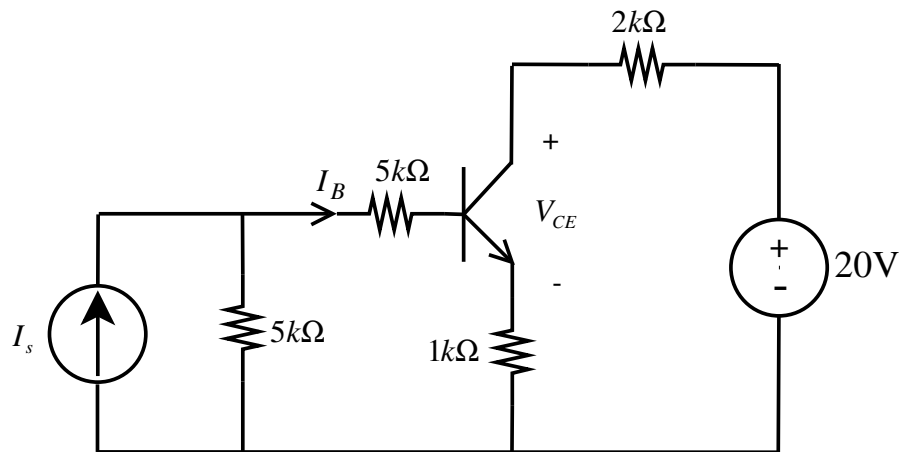


Bereken die Norton ekwivalent tov terminale $a-b$. [4]

Determine the Norton equivalent wrt terminals $a-b$. [4]

Vraag 2.9

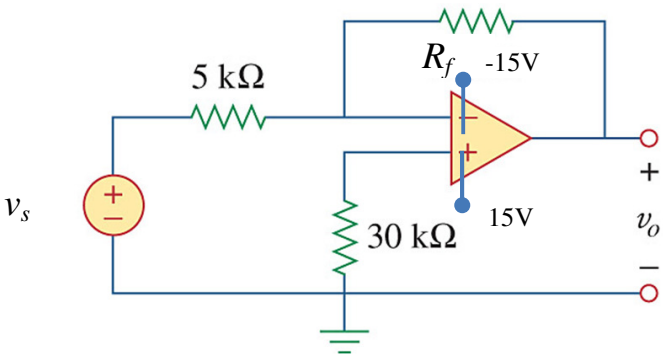
Question 2.9



- (a) Aanvaar $I_s = 1\text{mA}$ en bepaal I_B in die stroombaan ($V_{BE} = 0.7\text{V}$ en $\beta = 100$). [2]
 (b) Aanvaar $I_B = 30\mu\text{A}$ en bepaal V_{CE} in die stroombaan ($V_{BE} = 0.7\text{V}$ en $\beta = 100$). [2]

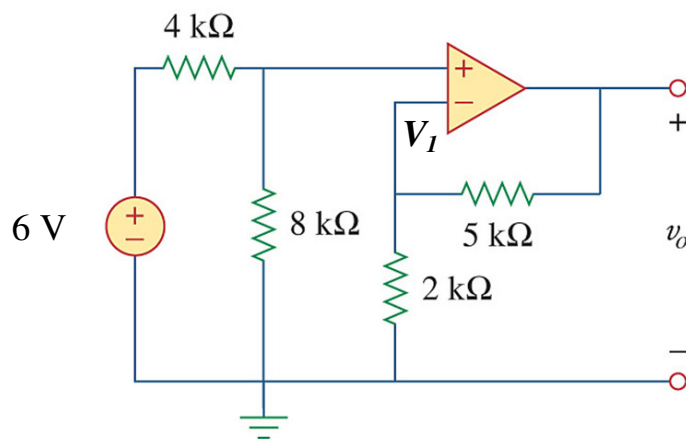
- (a) Assume $I_s = 1\text{mA}$ and determine I_B in the circuit ($V_{BE} = 0.7\text{V}$ and $\beta = 100$). [2]
 (b) Assume $I_B = 30\mu\text{A}$ and determine V_{CE} in the circuit ($V_{BE} = 0.7\text{V}$ and $\beta = 100$). [2]

VRAAG 3 / QUESTION 3 [13]

Vraag 3.1	Question 3.1
	
a) Bereken R_f sodat $v_o = -12\text{ V}$ vir $v_s = 2\text{ V}$. [1] b) Indien $R_f = 50\text{ k}\Omega$, vir watter waarde(s) van v_s sal die OV versadig? [2]	a) Calculate R_f so that $v_o = -12\text{ V}$ for $v_s = 2\text{ V}$. [1] b) If $R_f = 50\text{ k}\Omega$, at what value(s) for v_s will the opamp saturate? [2]

Vraag 3.2

Question 3.2

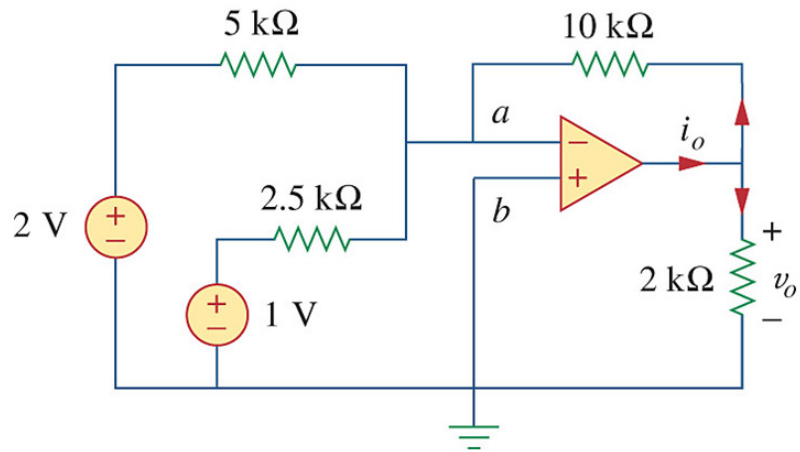


- a) Bereken V_I [1]
b) Bereken v_o [2]

- a) Calculate V_I [1]
b) Calculate v_o [2]

Vraag 3.3

Question 3.3



Bepaal:

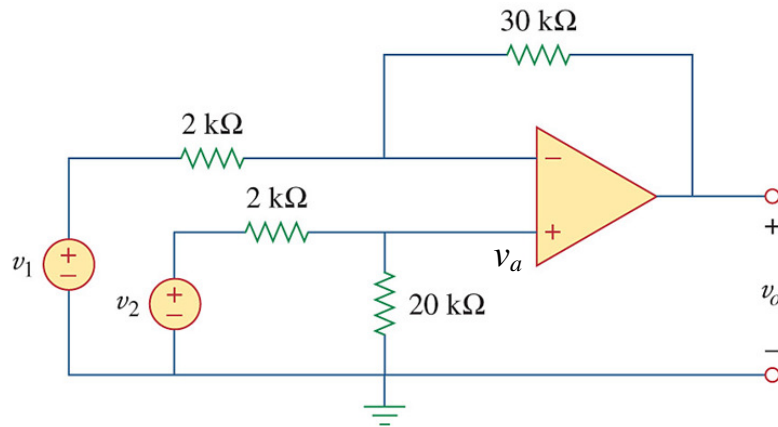
- a) $v_o[1]$
- b) $i_o[1]$

Determine

- a) $v_o[1]$
- b) $i_o[1]$

Vraag 3.4

Question 3.4

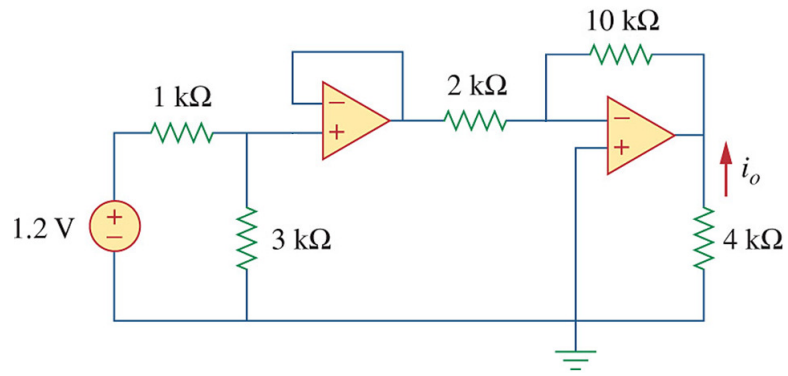


- a) Bepaal v_a indien $v_1 = 1$ V en v_o gemeet word as 14.1 V. [2]
 b) Bepaal v_2 [1]

- a) Determine v_a if $v_1 = 1$ V and v_o is measured as 14.1 V. [2]
 b) Determine v_2 [1]

Vraag 3.5

Question 3.5



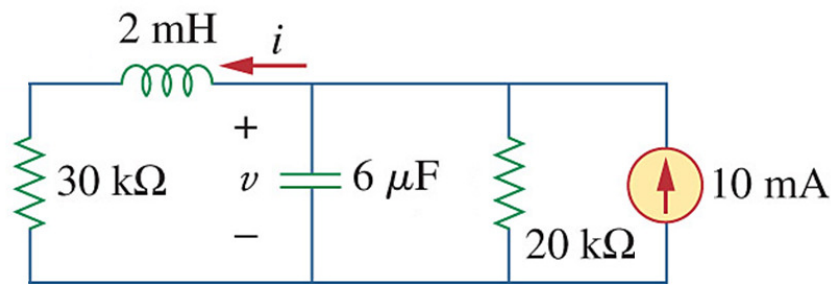
Bepaal i_o . [2]

Determine i_o . [2]

VRAAG 4 / QUESTION 4 [6]

Vraag 4.1	Question 4.1
Die stroom deur 'n 2 mH induktor is: $i(t) = 10 \cos 50t$ mA. Bereken a) Die spanning oor die induktor [1] b) Die energie gestoor in die induktor. [1] LW: Eenhede!	The current through a 2 mH inductor is: $i(t) = 10 \cos 50t$ mA. Calculate: a) The voltage across the inductor. [1] b) The energy stored in the inductor. [1] NB: Units!

Vraag 4.2	Question 4.2
The circuit diagram shows a network of capacitors. On the left, there are two open terminals. A 4 F capacitor is connected in series between these terminals. After the 4 F capacitor, the circuit splits into two parallel branches. The first branch contains a 3 F capacitor. The second branch contains a 12 F capacitor in series with a 4 F capacitor. These two branches recombine, and a 6 F capacitor is connected in parallel across the entire network. The output terminals are on the right.	
Bepaal die ekwivalente kapasitansie. [1]	Determine the equivalent capacitance. [1]

Vraag 4.3
Question 4.3


Onder bestendige gs toetstande, bepaal:

- a) Die stroom i . [1]
- b) Die spanning v . [1]
- c) Die energie gestoor in die kapasitor. [1]

Under steady-state dc conditions, determine:

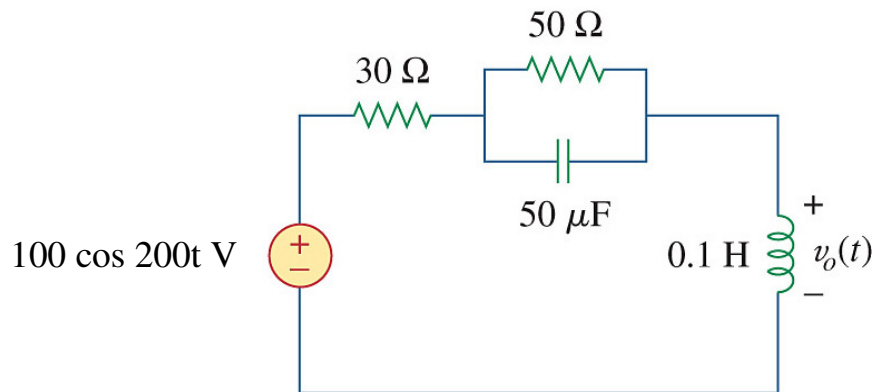
- a) The current i . [1]
- b) The voltage v . [1]
- c) The energy stored in the capacitor. [1]

VRAAG 5 / QUESTION 5 [12]

Vraag 5.1	Question 5.1
a) Skryf as 'n enkele cosinus uitdrukking: $i(t) = 20\cos(\omega t - 50^\circ) + 30\cos(\omega t + 60^\circ)$ A [2] b) Bepaal die ooreenstemmende fasor (in polêre vorm): $v(t) = -40\cos(50t + 35^\circ)$ V [1] c) Bepaal die ooreenstemmende fasor (in polêre vorm): $i(t) = 5\sin(15t + 60^\circ)$ A [1] d) Watter van die volgende fasors is loop voor? $V_1 = 25\angle 190^\circ$, $V_2 = 10\angle -20^\circ$ [1] e) Evalueer die determinant: [2] $\begin{vmatrix} 5 + j3 & 4 - 5j \\ -2 & -3 + 2j \end{vmatrix}$	a) Write as a single cosine expression: $i(t) = 20\cos(\omega t - 50^\circ) + 30\cos(\omega t + 60^\circ)$ A [2] b) Determine the corresponding phasor (in polar form): $v(t) = -40\cos(50t + 35^\circ)$ V [1] c) Determine the corresponding phasor (in polar form): $i(t) = 5\sin(15t + 60^\circ)$ A [1] d) Which of the following phasors is leading? $V_1 = 25\angle 190^\circ$, $V_2 = 10\angle -20^\circ$ [1] e) Evaluate the determinant: [2] $\begin{vmatrix} 5 + j3 & 4 - 5j \\ -2 & -3 + 2j \end{vmatrix}$

Vraag 5.2

Question 5.2

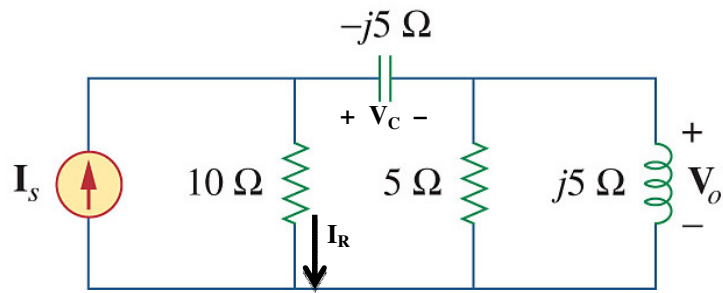


Bepaal $v_o(t)$. [3]

Determine $v_o(t)$. [3]

Vraag 5.3

Question 5.3



Indien $V_o = 20\angle 45^\circ$ V en $I_R = 2\angle -45^\circ$ A, bepaal

a) V_C [1]

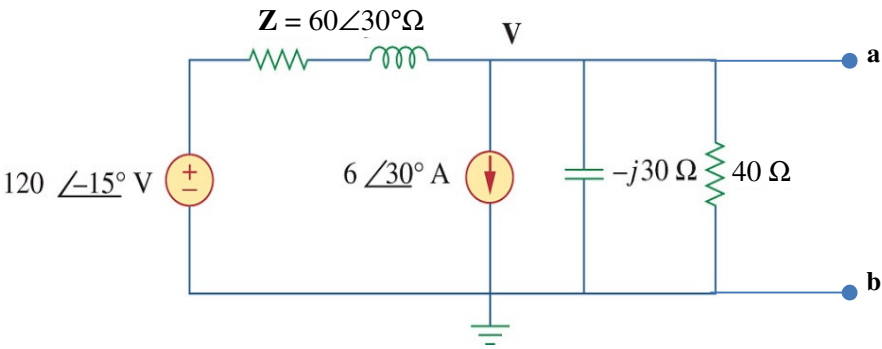
b) I_s [1]

If $V_o = 20\angle 45^\circ$ V and $I_R = 2\angle -45^\circ$ A, determine

a) V_C [1]

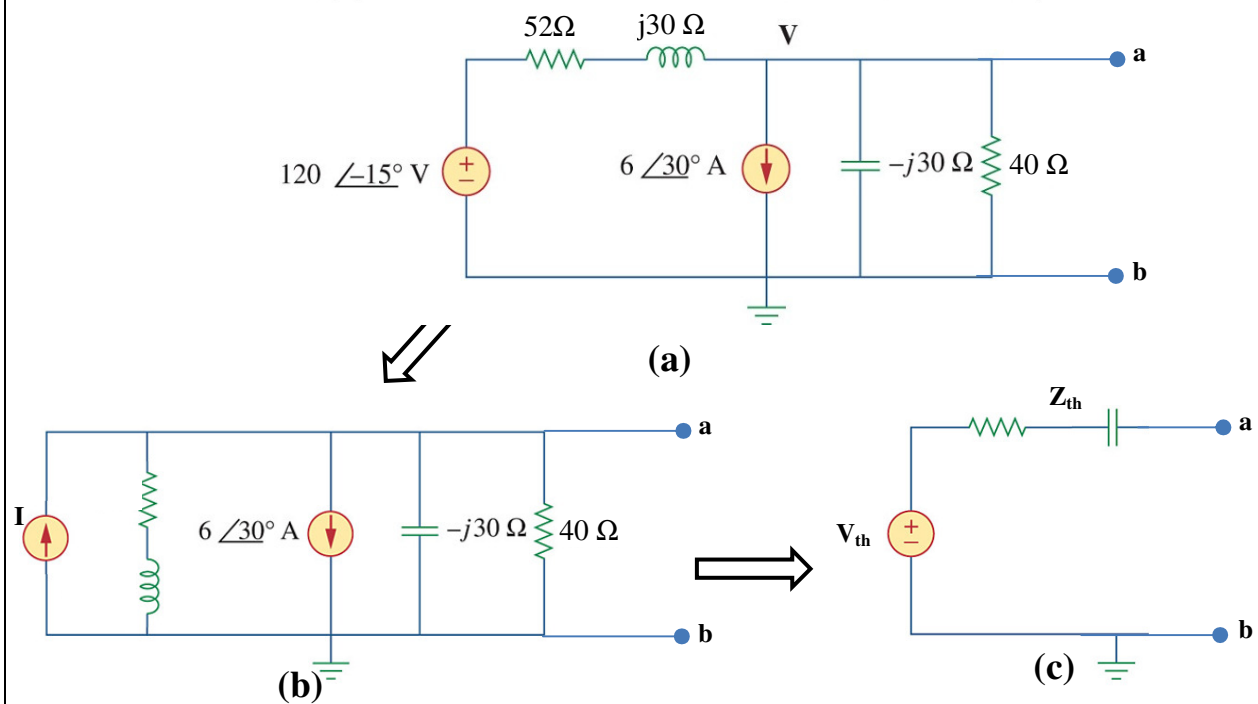
b) I_s [1]

VRAAG 6 / QUESTION 6 [14]

Vraag 6.1	Question 6.1
	
a) Skryf KCL neer by node a in die vorm: $term1 + term2 + \dots = 0$. Moenie vereenvoudig nie! [1] b) Bepaal nou die Thevenin spanning V_{th} tussen nodes a en b. [2]	a) Write down KCL at node a in the form: $term1 + term2 + \dots = 0$. Do not simplify! [1] b) Now determine the thevenin voltage V_{th} between nodes a and b. [2]

Vraag 6.2

Question 6.2



Bepaal die volgende dmv brontransformatie:

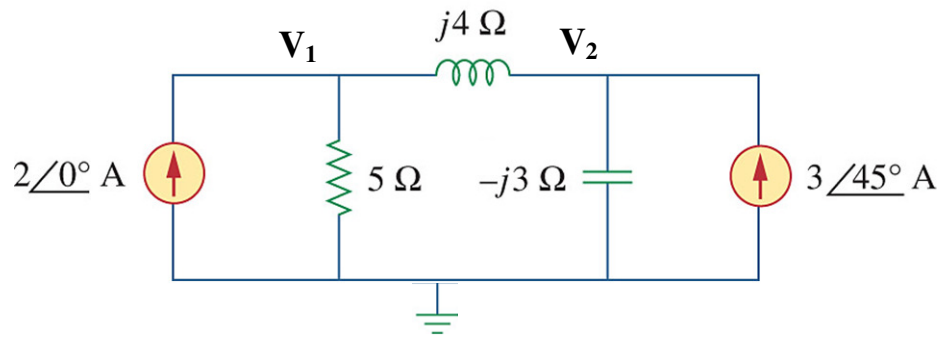
- $\mathbf{I}$ in figuur (b) in polar form.[1]
- $\mathbf{Z}_{th}$ in figuur (c) in polar form.[1]
- $\mathbf{V}_{th}$ in figuur (c) in polar form.[1]

Use source transformation to determine:

- $\mathbf{I}$ in figure (b) in polêre vorm.[1]
- $\mathbf{Z}_{th}$ in figure (c) in polêre vorm.[1]
- $\mathbf{V}_{th}$ in figure (c) in polêre vorm.[1]

Vraag 6.3

Question 6.3



Stel die node-analise vergelykings op vir V_1 en V_2 . Skryf dit in die volgende vorm:

$$(a_1 + jb_1)V_1 + j5V_2 = A_2\angle\phi_2 [2]$$

$$3V_1 + (a_3 + jb_3)V_2 = A_4\angle\phi_4 [2]$$

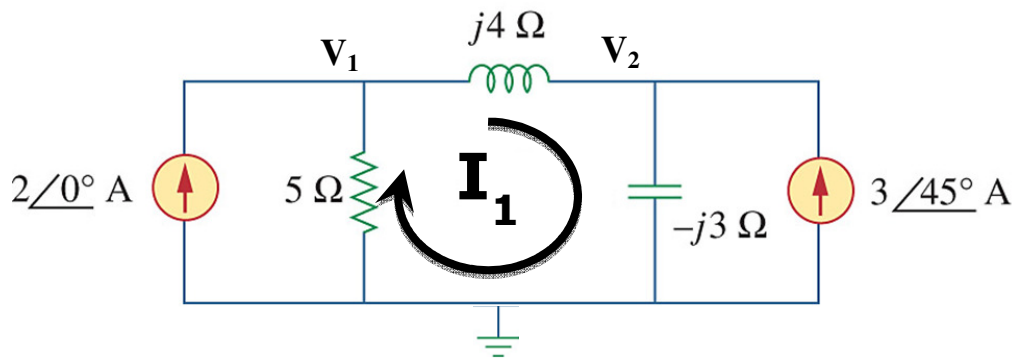
Set up the nodal analysis equations for V_1 and V_2 . Write it in the following form:

$$(a_1 + jb_1)V_1 + j5V_2 = A_2\angle\phi_2 [2]$$

$$3V_1 + (a_3 + jb_3)V_2 = A_4\angle\phi_4 [2]$$

Vraag 6.4

Question 6.4



a) Bepaal die lus-stroom $\mathbf{I}_1$. [2]

b) Gestel dat $\mathbf{I}_1 = 1.5\angle 50^\circ$ A, (wat dit nie is nie), bepaal $\mathbf{V}_1$ en $\mathbf{V}_2$. [2]

All answers must be in **polar form**.

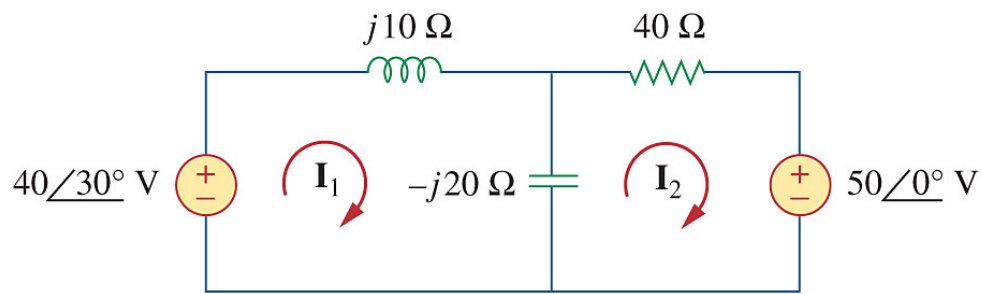
a) Determine the mesh current $\mathbf{I}_1$. [2]

b) Assuming $\mathbf{I}_1 = 1.5\angle 50^\circ$ A, (which it isn't), determine $\mathbf{V}_1$ and $\mathbf{V}_2$. [2]

Alle antwoorde moet in **polêre vorm** wees.

Vraag 6.5

Question 6.5



Stel die node-analise vergelykings op vir $\mathbf{I_1}$ en $\mathbf{I_2}$. Skryf dit in die volgende vorm:

$$\begin{aligned}(x_1 + jy_1)\mathbf{I_1} + j20\mathbf{I_2} &= X_2\angle\phi_2 [2] \\ j20\mathbf{I_1} + (x_3 + jy_3)\mathbf{I_2} &= X_4\angle\phi_4 [2]\end{aligned}$$

Set up the mesh analysis equations for $\mathbf{I_1}$ and $\mathbf{I_2}$. Write it in the following form:

$$\begin{aligned}(x_1 + jy_1)\mathbf{I_1} + j20\mathbf{I_2} &= X_2\angle\phi_2 [2] \\ j20\mathbf{I_1} + (x_3 + jy_3)\mathbf{I_2} &= X_4\angle\phi_4 [2]\end{aligned}$$

EBN122 EKSAMEN ANTWOORDBLAD/ EXAM ANSWER SHEET

14 NOVEMBER 2011

Student se besonderhede/ <i>Student's details</i>									
Van en voorletters: <i>Surname and initials:</i>					Studentenommer: <i>Student number:</i>				
Graadkursus: <i>Degree Course:</i>									
Tel. nr.: <i>Tel no.:</i>									

PUNTE/MARKS							
Q1	/11	Q2	/34	Q3	/13	Final	/90
Q4	/6	Q5	/12	Q6	/14		

VRAAG /QUESTION 1 [11]	
1.1	$p(1ms) =$
	$W =$
1.2	$P1 =$
	$P2 =$
	$P3 =$
1.3	$i_1 =$
	$i_2 =$
1.4	$R_{ab} =$

VRAAG /QUESTION 2 [34]		
2.1	$a =$	$b =$
	$c =$	$d =$
2.2	$a =$	$b =$
	$c =$	$d =$
2.3	$V_1 =$	
	$i_2 =$	
2.4	$V_s =$	$R =$
2.5	$i_{x(7V)} =$	$i_{x(3A)} =$
2.6	$V_x =$	
2.7	$V_{th} =$	$R_{th} =$
2.8	$I_N =$	$R_N =$
2.9	$I_B =$	$V_{CE} =$

VRAAG 3 / QUESTION 3 [13]

3.1	a) $R_f =$ [1]	b) $v_s =$ [2]		3.4	a) $v_a =$ [2]	b) $v_2 =$ [1]
3.2	a) $V_f =$ [1]	b) $v_o =$ [2]		3.5	$i_o =$ [2]	
3.3	a) $v_o =$ [1]	b) $i_o =$ [1]				

VRAAG 4 / QUESTION 4 [6]

4.1	a) $v(t) =$ [1]			4.3	a) $i =$ [1]
	b) $w(t) =$ [1]				b) $v =$ [1]
4.2	$C_{eq} =$ [1]				c) $w =$ [1]

VRAAG 5 / QUESTION 5 [12]

Vraag 5.1 / Question 5.1			
a [2]		5.2	$v_o(t) =$ [3]
b [1]		5.3	a) $V_C =$ [1]
c [1]			b) $I_s =$ [1]
d [1]			
e [2]			

VRAAG 6/ QUESTION 6 [14]

6.1a [1]			
6.1b	$V =$ [2]		
6.2	$I =$ [1]	$Z =$ [1]	$V =$ [1]
6.3	$(a_1 + jb_1) =$		$A_2 \angle \phi_2^\circ =$
	$(a_3 + jb_3) =$		$A_4 \angle \phi_4^\circ =$
6.4	$I_1 =$	$V_1 =$	$V_2 =$
6.5	$(x_1 + jy_1) =$		$X_2 \angle \phi_2^\circ =$
	$(x_3 + jy_3) =$		$X_4 \angle \phi_4^\circ =$